CIA - 1

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2. Given:
Joint distribution $P(X,Y)$ of load X (low, medium, high) and fault Y (No, yes); model $Q = (0.5, 0.3, 0.2)$.
All logarithms are base 2.

Load X	Y = No	Y = Yes	P(X)
Low	0.30	0.02	0.32
Medium	0.28	0.10	0.38
High	0.10	0.20	0.30
P(Y)	0.68	0.32	1.00

1. Marginals: $P(X) = (0.32, 0.38, 0.30)$; $P(Y) = (0.68, 0.32)$
Row & column sums of the joint table

2. $H(X) = -\sum p \log_2 p$
$= 0.5260 + 0.5304 + 0.5211 = 1.5776$ bits

$H(Y) = 0.3784 + 0.5260 = 0.9044$ bits

Each term is $-p \log_2 p$

Eg. $-0.32 \log_2 0.32 = 0.5260$

3. $H(X,Y) = 0.5211 + 0.1129 + 0.5142 + 0.3322 + 0.3322 + 0.4644$
$= 2.2770$ bits

Sum over all six joint probabilities

4. $H(Y|X) = H(X,Y) - H(X) = 2.2770 - 1.5776$
$= 0.6994$ bits

chain rule of entropy.

5. $I(X;Y) = H(Y) - H(Y|X) = 0.9044 - 0.6994 = 0.2050$ bits

$I(X;Y)$ is 22.7% of $H(Y)$

6. cross-entropy $H(P,Q) = -\sum P \log Q = 0.32(1.0000) + 0.38(1.7370) + 0.30(2.3219) = 1.6766$ bits

7. KL divergence $D(P||Q) = H(P,Q) - H(X)$

$= 1.6766 - 1.5776$

$= 0.0990$ bits

8. $P(Y=\text{yes} / X=\text{high}) = 0.20/0.30 = 0.6667$

Interpretation & conclusion:

the mutual information of 0.205 is +ve, Model Q has $D_{KL} = 0.099$ bits; it over-weights low load (0.5 vs 0.32) & under weights high load, so it is only a coarse approx & should be re-estimated from observed frequencies

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4. Bayes decision theory in Hospital Screening:

given:

prevalence: $P(C) = 0.03$, $P(\bar{C}) = 0.97$

Sensitivity = 90%, Specificity = 94%.

cost of missing disease = 50 units

Cost of treating healthy patient = 2 units

i) probability of positive test

$$P(+) = P(+/C)\,P(C) + P(+/\bar{C})\,P(\bar{C})$$

$= 0.9(0.03) + 0.06(0.97)$

$P(+) = 0.08852 = 8.52\%$

posterior probabilities using Bayes theorem:

$$P(C/+) = \frac{0.9(0.03)}{0.0852} = \boxed{0.3169}$$

$$P(C/-) = \frac{0.1(0.03)}{0.9148} = \boxed{0.00328}$$

3. Conditional risks & decisions

Risk of treating = $2P(\bar{C}/x)$

Risk of not treating = $50P(C/x)$

outcome	treat risk	No-treatment risk	decision
positive	1.366	15.845	Treat
negative	1.993	0.164	Do not treat

4. Overall Bayes risk

$$R = 2P(\bar{C}, +) + 50P(C, -)$$

$$= 2(0.0582) + 50(0.003)$$

$$\boxed{R = 0.2664}$$

5. Two independent positive tests

$$P(C/+,+) = \frac{0.03(0.9)^2}{0.03(0.9)^2 + 0.97(0.06)^2}$$

$$\boxed{P(C/+,+) = 87.44\%}$$

Conclusion:

Minimum-risk decisions minimise expected cost. unlike the minimum error rule. It is more suitable for hospital screening because missing disease is expensive

6. Bernoulli Naive Bayes spam classification:

given 100 emails → 40 spam & 60 legitimate

a) class priors & smoothed probabilities

$P(S) = \frac{40}{100} = 0.4$, $P(L) = 0.6$

Laplace smoothing $\boxed{P(w/c) = \frac{n_{wc}+1}{N_c+2}}$

Word	spam	legitimate
offer	29/42 = 0.6905	7/62 = 0.1129
free	25/42 = 0.5952	4/62 = 0.0645
meeting	5/42 = 0.1190	34/62 = 0.5484

(b) Classify the email

$S_S = 0.4\left(\frac{29}{42}\right)\left(\frac{25}{42}\right)\left(\frac{37}{42}\right)$

$\boxed{S_S = 0.14483}$

legitimate score $S_L = 0.6\left(\frac{7}{62}\right)\left(\frac{4}{62}\right)\left(\frac{38}{62}\right)$

$\boxed{S_L = 0.001974}$

$P(S/x) = \frac{S_S}{S_S + S_L} \Rightarrow$ $\boxed{P(S/x) = 98.66\%}$

(c) cost sensitive decision

False positive cost $= \frac{5}{5+1} = 0.8333$

$0.9866 > 0.833$

⇒ classify the email as spam

(d) performance measures

given TP = 45, FN = 15, FP = 100

$\boxed{\text{Accuracy} = \frac{45+130}{200} \times 100 = 87.5\%}$

$\boxed{\text{precision} = \frac{45}{45+10} \times 100 = 81.82\%}$

Recall = $\frac{45}{45+15} \times 100 = 75\%$.

$F1 = \frac{2PR}{P+R} = 78.26\%$.

Total misclassification cost: $5(FP) + FN = 5(10) + 15 = 65$ units

Conclusion:
Classifier achieves 87.5% accuracy misses 25% spam emails

8. Back-propagation in a neural network

Given:

$x_1 = 0.5$, $x_2 = 1$, $t = 1$, $\eta = 0.5$

Sigmoid function $\sigma(z) = \frac{1}{1+e^{-z}}$

1. Hidden unit activations

for h_1: net $h_1 = 0.4(0.5) - 0.2(1) + 0.1 = 0.1$

$h_1 = 0.5250$

for h_2: net $h_2 = 0.3(0.5) + 0.5(1) - 0.1 = 0.5$

$h_2 = 0.6341$

2 output & error

net $y = 0.6h_1 - 0.4h_2 + 0.2 = 0.2613$

$y = \sigma(0.2613) = 0.5650$

$E = \frac{1}{2}(t-y)^2 = 0.09413$

3 Error terms

output unit $\delta y = (t-y)y(1-y)$ $\quad \delta y = 0.10692$

$\delta h_1 = h_1(1-h_1)v_1 \delta y = 0.01600$

$\delta h_2 = h_2(1-h_2)v_2 \delta y = -0.009923$

4 updated weights & biases

$w^{new} = w + \eta \delta x$

$b^{new} = b + \eta \delta$

5. Second forward pass

$y' \approx 0.58778$

$E \approx \frac{1}{2}(1-0.58778)^2 = 0.08496$

percentage error reduction $= \frac{0.09463 - 0.0849}{0.08496} \times 100$

$= 10.21\%$

Conclusion:

The error decreases by 10.21% showing successful learning. A high learning rate can cause unstable updates

10. Entropy, Mutual Information, KL divergence

given

Load X	Y = No	Y = yes
low	0.30	0.02
medium	0.28	0.10
high	0.10	0.20

1. Marginal distributions

For load : $P(X) = (0.32, 0.38, 0.30)$

for faults: $P(Y) = (0.68, 0.32)$

2 Entropy of load

$$H(X) = -\sum P(x) \log_2 P(x)$$

$H(X) = 1.5776$ bits

3 Entropy of faults

$H(Y) = -[0.68 \log_2(0.68) + 0.32 \log_2(0.32)]$

$= 0.9044$ bits

4. Joint entropy

$$H(X,Y) = -\sum_{x,y} P(x,y) \log_2 P(x,y)$$

Substituting:

$$H(X,Y) = 2.2770 \text{ bits}$$

5. Conditional entropy

$$H(Y|X) = H(X,Y) - H(X)$$
$$= 2.2770 - 1.5776$$
$$= 0.6994 \text{ bits}$$

6. Mutual Information

$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$
$$= 1.5776 + 0.9044 - 2.2770$$
$$I(X;Y) = 0.2050 \text{ bits}$$

Conclusion:

High load is associated with higher fault probability. Mutual Information shows limited predictive value